Multisets of $O(k)$-invariants under a labelling symmetry group

Working notes

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1 Setup

Let $L = (v_1, v_2, \ldots, v_n)$ be an ordered list of n vectors, each $v_i \in \mathbb{F}^k$, where $\mathbb{F}$ is usually $\mathbb{R}$ but may be $\mathbb{C}$.

1.1 Invariants

Definition 1 (Dots)**.** For $1 \le i \le j \le n$ let

$$d_{ij} = v_i \cdot v_j = v_i^{\mathsf{T}} v_j.$$

There are $n + \binom{n}{2}$ of these (self-pairs included). The product is bilinear (no complex conjugation) also when $\mathbb{F} = \mathbb{C}$.

Definition 2 (Epsilons)**.** For $1 \le i_1 < i_2 < \cdots < i_k \le n$ let

$$e_{i_1 i_2 \cdots i_k} = \mathrm{Det}\big[v_{i_1}, v_{i_2}, \ldots, v_{i_k}\big].$$

There are $\binom{n}{k}$ of these. The inequalities are strict: any repeated index gives a zero determinant.

For non-canonically-ordered indices we use $d_{ji} = d_{ij}$ and $e_{\ldots j \ldots i \ldots} = -e_{\ldots i \ldots j \ldots}$ (total antisymmetry), so any index list can be brought to canonical (increasing) order, gaining a sign $\mathrm{sgn}(\pi)$ for the sorting permutation π in the case of an e.

1.2 The vector w

Let w be the column vector containing every d and every e in a fixed, predetermined order. Any order is allowed, but it is convenient to keep together the entries that form a single orbit under the permutation part of the group G (below).

1.3 The symmetry group G

G is generated by

- a subgroup of S_n acting on the labels $1, \ldots, n$, typically a product $S(n_1) \times S(n_2) \times \cdots$ permuting labels only within blocks. For example, for $n = 5$, $S(3) \times S(2)$ permutes $\{v_1, v_2, v_3\}$ among themselves and $\{v_4, v_5\}$ among themselves;
- optionally, a parity factor $P = \{1, p\}$, so that $|G \times P| = 2|G|$.

Definition 3 (Parity)**.** p acts on every vector by the same fixed reflection,

$$p : v_i \mapsto R\, v_i, \qquad R = \mathrm{diag}(1, \ldots, 1, -1),$$

that is, it flips the sign of the last component of every vector.

Lemma 1. *For every k, p leaves each d unchanged and negates each e:*

$$p : d_{ij} \mapsto d_{ij}, \qquad p : e_{i_1 \cdots i_k} \mapsto -e_{i_1 \cdots i_k}.$$

Proof. $R^{\mathsf{T}}R = I$, so $(Rv_i)^{\mathsf{T}}(Rv_j) = v_i^{\mathsf{T}}v_j$. Also $\mathrm{Det}[Rv_{i_1}, \ldots, Rv_{i_k}] = \det(R)\,\mathrm{Det}[v_{i_1}, \ldots, v_{i_k}]$ and $\det R = -1$. □

Remark 1. The proof uses only $R^{\mathsf{T}}R = I$ and $\det R = -1$. So any reflection, or more generally any orthogonal matrix with $\det = -1$, acts identically on w, and the choice of which component to flip is immaterial. Because R is real, the d's are invariant under either a bilinear or a Hermitian product. The e's, however, pair naturally with the bilinear d.

Remark 2 (Point inversion is *not* the right parity). The alternative $v \mapsto -v$ (that is, $R = -I_k$) has $\det(-I_k) = (-1)^k$. It negates the e's only for odd k. For even k (for example $k = 2$, where $-I_2$ is a rotation by π) it leaves w completely unchanged. This is why parity is defined as a reflection above.

1.4 Group action on w and the multiset

For $g \in G$, the vector $g \cdot w$ is obtained as follows:

1. relabel the indices of each entry according to the permutation part of g;
2. restore canonical index order, using $d_{ji} = d_{ij}$ and the antisymmetry sign for each e;
3. if g contains p, negate every e (Lemma 1).

The resulting entries are re-expressed in terms of the original (canonical) d's and e's, keeping the slot positions of w fixed. Define the multiset

$$\mathrm{MS} = \{\, g \cdot w \mid g \in G \,\},$$

with multiplicities, so that $|\mathrm{MS}| = |G|$ counted with multiplicity.

2 Worked example: $k = 2$, $L = (a, b, p)$

Here $n = 3$. To avoid a clash with the parity element, the vector named p in L is written as a subscript only. There are $3 + 3 = 6$ dots and $\binom{3}{2} = 3$ epsilons. Ordered by orbits of $S(2) \times S(1)$ (bars separate the orbits):

$$w = \big(\, d_{aa}, d_{bb} \;\big|\; d_{pp} \;\big|\; d_{ab} \;\big|\; d_{ap}, d_{bp} \;\big|\; e_{ab} \;\big|\; e_{ap}, e_{bp} \,\big)^{\mathsf{T}}.$$

Let $\sigma = (ab)$. Under σ: $d_{ab} \mapsto d_{ba} = d_{ab}$, $e_{ab} \mapsto e_{ba} = -e_{ab}$, and $e_{ap} \leftrightarrow e_{bp}$ with no sign change.

2.1 $G = S(2) \times S(1)$

$$\mathrm{MS} = \left\{ \underbrace{\begin{pmatrix} d_{aa} \\ d_{bb} \\ d_{pp} \\ d_{ab} \\ d_{ap} \\ d_{bp} \\ e_{ab} \\ e_{ap} \\ e_{bp} \end{pmatrix}}_{1}, \underbrace{\begin{pmatrix} d_{bb} \\ d_{aa} \\ d_{pp} \\ d_{ab} \\ d_{bp} \\ d_{ap} \\ -e_{ab} \\ e_{bp} \\ e_{ap} \end{pmatrix}}_{\sigma} \right\}$$

2.2 $G = S(2) \times S(1) \times P$

$$\mathrm{MS} = \left\{ \underbrace{\begin{pmatrix} d_{aa} \\ d_{bb} \\ d_{pp} \\ d_{ab} \\ d_{ap} \\ d_{bp} \\ e_{ab} \\ e_{ap} \\ e_{bp} \end{pmatrix}}_{1}, \underbrace{\begin{pmatrix} d_{bb} \\ d_{aa} \\ d_{pp} \\ d_{ab} \\ d_{bp} \\ d_{ap} \\ -e_{ab} \\ e_{bp} \\ e_{ap} \end{pmatrix}}_{\sigma}, \underbrace{\begin{pmatrix} d_{aa} \\ d_{bb} \\ d_{pp} \\ d_{ab} \\ d_{ap} \\ d_{bp} \\ -e_{ab} \\ -e_{ap} \\ -e_{bp} \end{pmatrix}}_{p}, \underbrace{\begin{pmatrix} d_{bb} \\ d_{aa} \\ d_{pp} \\ d_{ab} \\ d_{bp} \\ d_{ap} \\ e_{ab} \\ -e_{bp} \\ -e_{ap} \end{pmatrix}}_{\sigma p} \right\}$$

In σp, the swap sign and the parity sign cancel on e_{ab}.

3 Layer 2: embedding the multiset

The multiset MS is not yet a usable output. Next we apply a continuous, injective and algebraic *multiset embedding*

$$\Phi : \big(\mathbb{F}^m\big)^{|G|}/S_{|G|} \longrightarrow \mathbb{F}^D, \qquad m = \dim w,$$

and express each component of $\Phi(\mathrm{MS})$ explicitly as a polynomial in the d's and e's. Several choices of Φ are possible. We start with the Bursarial embedding because its outputs are explicit low-degree polynomials, which makes it the easiest to work with.

3.1 The Bursarial embedding

Index the slots of w from 0 to $m-1$, so $w = (w_0, \ldots, w_{m-1})^{\mathsf{T}}$. For each $g \in G$ define

$$q_g(x) = \sum_{j=0}^{m-1} (g \cdot w)_j \, x^j,$$

and multiply one linear factor in y for each group element:

$$B(x, y) = \prod_{g \in G} \big(y + q_g(x)\big).$$

Elementary symmetric polynomials. For N quantities $z_1, \ldots, z_N$, the *c-th elementary symmetric polynomial* is the sum of all products of c of them with distinct indices:

$$E_c(z_1, \ldots, z_N) = \sum_{1 \le i_1 < i_2 < \cdots < i_c \le N} z_{i_1} z_{i_2} \cdots z_{i_c}, \qquad E_0 = 1.$$

For example, $E_1 = z_1 + \cdots + z_N$ and $E_N = z_1 z_2 \cdots z_N$. When $N = 3$, $E_2 = z_1 z_2 + z_1 z_3 + z_2 z_3$. They are the coefficients that appear when a product of linear factors is multiplied out:

$$\prod_{i=1}^{N} (y + z_i) = \sum_{c=0}^{N} y^{N-c} \, E_c(z_1, \ldots, z_N).$$

(For $N = 2$ this is $(y + z_1)(y + z_2) = y^2 + y(z_1 + z_2) + z_1 z_2$.)

Applying this to B. Take $N = |G|$ and let the z's be the polynomials $q_g(x)$, one for each $g \in G$. Write

$$E_c(x) \equiv E_c\big(\{q_g(x)\}_{g\in G}\big) = \sum_{\substack{\{g_1,\dots,g_c\}\subseteq G \\ (c \text{ distinct elements})}} q_{g_1}(x)\, q_{g_2}(x) \cdots q_{g_c}(x).$$

Then

$$B(x,y) = \sum_{c=0}^{|G|} y^{|G|-c}\, E_c(x).$$

Each q_g has degree (at most) $m-1$ in x, so $E_c(x)$, a sum of products of c of them, is a polynomial in x of degree (at most) $c(m-1)$. Write its coefficients as $E_{c,j}$:

$$E_c(x) = \sum_{j=0}^{c(m-1)} E_{c,j}\, x^j.$$

(The notation $[x^j]\, f$ is sometimes used for the coefficient of x^j in f, so that $E_{c,j} = [x^j]\, E_c$.) The leading term $y^{|G|}$ ($c = 0$) carries no information. The embedding is the list of all the remaining coefficients:

$$\Phi(\mathrm{MS}) = \big(E_{c,j}\big)_{1\le c\le |G|,\ 0\le j\le c(m-1)},$$

which has

$$D = \sum_{c=1}^{|G|} \big(c(m-1)+1\big) = |G| + (m-1)\,\frac{|G|(|G|+1)}{2}$$

components. This is the construction used in Echinocoder's `Cinf_sympy_bursar_embedder_for_array_of_reals_as_multiset.py`, with $n \to |G|$ and $k \to m$.

Properties.

- *Injective on multisets.* $\mathbb{F}[x][y]$ is a unique factorisation domain, and each factor $y + q_g(x)$ is monic in y. So B determines the multiset $\{q_g\}$, and hence MS.
- *Invariant.* B is a product over all of G, so $\Phi(\mathrm{MS})$ is unchanged when L is acted on by any $g \in G$.
- *Polynomial.* $E_{c,j}$ is a homogeneous polynomial of degree c in the entries of w, that is, in the d's and e's:

$$E_{c,j} = \sum_{\{g_1,\dots,g_c\}\subseteq G} \ \sum_{j_1+\cdots+j_c=j} \ \prod_{r=1}^{c} (g_r \cdot w)_{j_r}.$$

- *Order-dependent coordinates.* The slot order of w decides which entries multiply which powers of x. Changing it changes the coordinates of Φ, but not its injectivity.

What injectivity is actually required. We never want to recover L itself. The purpose of MS is to forget exactly the information that G moves. Using d's and e's erases the orientation of L in $\mathbb{F}^k$, that is, the action of $SO(k)$. Over $\mathbb{R}$ the Gram matrix (d_{ij}) determines L up to $O(k)$. The e's then fix the handedness whenever the v_i span $\mathbb{F}^k$; when they do not, there is no handedness to fix. Each factor of G erases something further. For example, $S(3)$ erases which of a, b, c is which, and P, if present, erases handedness, so that L is then known only up to $O(k)$. In that case the e's carry nothing beyond the d's. Every G-invariant is then even in the e's, and by Cauchy–Binet $e_I\, e_J = \det\big[d_{i_r j_s}\big]_{r,s=1}^{k}$. They are kept in w here only for uniformity. A practical implementation would leave them out whenever $P \subseteq G$, because they greatly enlarge

MS and everything that then has to be pruned. (Over $\mathbb{C}$ with a bilinear product the degenerate cases, such as nonzero v with $v \cdot v = 0$, still need checking.) Two notions of injectivity should therefore be kept apart:

(A) *Formal injectivity.* Treat the m entries of w as independent variables, and require Φ to separate $\{g \cdot w\}$ for *every* $w \in \mathbb{F}^m$. This is all the Bursarial embedding knows about.

(B) *Injectivity on the multisets that actually occur.* Require

$$\Phi\big(\mathrm{MS}(L)\big) = \Phi\big(\mathrm{MS}(L')\big) \implies \mathrm{MS}(L) = \mathrm{MS}(L'),$$

equivalently $L' = g \cdot (h\, L)$ for some $g \in G$ and orthogonal h (special orthogonal if $P \not\subseteq G$). Only those $w = w(L)$ lying on the variety cut out by the Gram and Plücker relations need to be separated.

(B) is the target. (A) implies (B), but (B) asks less of Φ, so it should be achievable with fewer outputs.

Redundancy. Φ is deliberately oversized, for two separate reasons. First, it does not know that MS is structured. MS is the G-orbit of a single point w, and each slot of $g \cdot w$ is $\pm$ an entry from the same G-orbit block of w, so many coefficients vanish identically and many others are functionally dependent on the rest. Second, it does not know the Gram and Plücker relations among the d's and e's, which are what separate target (B) from target (A). Exploiting both kinds of redundancy is the next layer. The present goal is only the oversized expressions, which are certainly injective, generated algorithmically and verifiably for general G.

3.2 Example: $k = 2$, $L = (a, b, p)$, $G = S(2) \times S(1)$

With the slot order of Section 2, $m = 9$, $|G| = 2$ and $D = 2 + 8 \cdot 3 = 26$:

$$q_1(x) = d_{aa} + d_{bb}x + d_{pp}x^2 + d_{ab}x^3 + d_{ap}x^4 + d_{bp}x^5 + e_{ab}x^6 + e_{ap}x^7 + e_{bp}x^8,$$
$$q_\sigma(x) = d_{bb} + d_{aa}x + d_{pp}x^2 + d_{ab}x^3 + d_{bp}x^4 + d_{ap}x^5 - e_{ab}x^6 + e_{bp}x^7 + e_{ap}x^8.$$

The $c = 1$ block ($E_1 = q_1 + q_\sigma$, 9 coefficients $E_{1,0}, \ldots, E_{1,8}$) is

$$\big(\, d_{aa}{+}d_{bb},\ d_{aa}{+}d_{bb},\ 2d_{pp},\ 2d_{ab},\ d_{ap}{+}d_{bp},\ d_{ap}{+}d_{bp},\ 0,\ e_{ap}{+}e_{bp},\ e_{ap}{+}e_{bp} \,\big),$$

which already shows a vanishing coefficient and repeated ones. The $c = 2$ block ($E_2 = q_1 q_\sigma$, 17 coefficients $E_{2,0}, \ldots, E_{2,16}$) contains, for example,

$$E_{2,12} = 2\big(d_{ap}e_{ap} + d_{bp}e_{bp}\big) - e_{ab}^2.$$

This is where the information about e_{ab} lost from E_1 reappears.

3.3 Involution packing

The Bursarial embedding can be made roughly half the size, while staying injective by construction (Level 0), whenever the multiset is invariant under a fixed linear involution. For MS this is always the case if G contains an element acting on w as a non-trivial involution.

The general mechanism. Let T be a real $m \times m$ matrix with $T^2 = I$, and let M be a multiset of vectors in $\mathbb{R}^m$ that is invariant under T as a multiset, that is, $\{Tv : v \in M\} = M$. Choose integer bases $f_1, \ldots, f_{m_+}$ and $h_1, \ldots, h_{m_-}$ of linear functionals with $f_a \circ T = f_a$ and $h_b \circ T = -h_b$. Here $m_\pm$ are the dimensions of the ± 1 eigenspaces, $m_+ + m_- = m$, and $m_+ - m_- = \operatorname{tr} T$. Pad the shorter list with zero functionals to a common length $r = \max(m_+, m_-) = (m + |\operatorname{tr} T|)/2$, and pack each vector as

$$\phi(v) = \big(f_1(v) + i\, h_1(v),\ \ldots,\ f_r(v) + i\, h_r(v)\big) \in \mathbb{C}^r.$$

Then ϕ is injective, and $\phi(Tv) = \overline{\phi(v)}$, entry by entry. With $q_v(x) = \sum_j \phi(v)_j\, x^j$, the complex Bursarial polynomial

$$B(x, y) = \prod_{v \in M} \big(y + q_v(x)\big)$$

therefore satisfies

$$\overline{B} = \prod_{v \in M} \big(y + q_{Tv}(x)\big) = B,$$

because $v \mapsto Tv$ permutes M. So every coefficient of B is real. $\mathbb{C}[x][y]$ is a unique factorisation domain and each factor is monic in y, so B determines the multiset $\{q_v\}$, hence $\{\phi(v)\}$, hence M. The real coefficients of B are therefore an injective embedding of M with

$$D = |M| + (r - 1)\, \frac{|M|(|M| + 1)}{2}$$

outputs, instead of $|M| + (m - 1)|M|(|M| + 1)/2$.

Application to MS**.** Let $g_0 \in G$ act on w as a non-trivial involution, and take T to be that action. Because $g \mapsto g_0 g$ permutes G,

$$T(g \cdot w) = (g_0 g) \cdot w,$$

so MS is invariant under T and the mechanism applies. The members of MS pair up as $(g \cdot w,\ g_0 g \cdot w)$. Note that g_0 multiplies on the *left*. Pairing $g \cdot w$ with $g g_0 \cdot w$ instead would relate the two by $g g_0 g^{-1}$, which depends on g and is not a single fixed T. When T is a signed permutation, $(Tw)_s = \sigma_s\, w_{\pi(s)}$, integer functionals can be read off slot by slot:

- a fixed slot ($\pi(s) = s$) gives w_s, in the $+1$ part if $\sigma_s = +1$ and in the -1 part otherwise;
- a swapped pair ($\pi(s) = t \neq s$) gives $w_s + \sigma_s w_t$ in the $+1$ part and $w_s - \sigma_s w_t$ in the -1 part.

The best g_0 is the one with smallest $|\operatorname{tr} T|$. The special case $g_0 = p$ (parity) packs every d with an e. The case of g_0 swapping the two members of a zipped pair is symcoder's $z \mapsto i\bar{z}$ compression.

Example. For $S(2)$ acting on (a, b) with $k = 2$, take $g_0 = (ab)$, which has trace 0. Then

$$z_1 = (d_{aa} + d_{bb}) + i\,(d_{aa} - d_{bb}), \qquad z_2 = d_{ab} + i\, e_{ab} = \bar{z}_a z_b,$$

and the 5 outputs are

$$2(d_{aa} + d_{bb}), \quad 2d_{ab}, \quad 2(d_{aa}^2 + d_{bb}^2), \quad 2\big(d_{ab}(d_{aa} + d_{bb}) + e_{ab}(d_{aa} - d_{bb})\big), \quad d_{ab}^2 + e_{ab}^2.$$

The unpacked embedding has 11.

Sizes. The table gives output counts before any reduction.

case	g_0	tr	plain	packed
$S(2)$, $k=2$	(ab)	0	11	5
$S(3)$, $k=2$	(bc)	+1	174	90
$S(2)\times S(1)$, $k=2$	(ab)	+1	26	14
$S(5)$, $k=4$	(de)	+4	138060	79980
$S(3)\times S(2)$, $k=3$	$(bc)(pq)$	+1	1884	948
$S(5)\times P$, $k=4$	$(bc)(de)p$	+2	549720	289440

Limits. The saving is at most a factor of about 2. Using two commuting involutions would need an algebra with two independent conjugations. Bicomplex numbers have zero divisors, which breaks unique factorisation, and quaternions do not commute, so the product depends on the order of its factors. On the simple cases of Section 6, full polynomial closure (in L) leaves exactly as many outputs from the packed embedding as from the plain one. The packing does not change what survives that (expensive) reduction. What it gives is the factor of 2 for free, with no reduction work, and it does better than the plain embedding followed by the linear (safe) reductions.

4 Layer 3: removing redundancy

There is no obviously best method for removing the redundancy in Φ. This section records what we are optimising and which reductions are safe.

4.1 Design objectives

1. *Injectivity on every input.* Target (B) must hold for every L, including degenerate ones: coincident or parallel vectors, and v_i that fail to span $\mathbb{F}^k$. Separation for generic L alone is not acceptable. We expect sets that separate everywhere to be strictly larger than sets that separate only generic L. The extra outputs are needed only to resolve confusions on degenerate inputs.

2. *Few outputs.* The number of outputs is to be minimised subject to (1). It can never be smaller than the dimension of the quotient, which for $n \geq k$ is $nk - \frac{1}{2}k(k-1)$. By Dym and Gortler, about twice the input dimension suffices if dense combinations are allowed, but see (3).

3. *Low complexity of each output, as a function of the raw inputs.* Separating invariants that have high degree and depend on delicate cancellations can be so "spiky" that, plotted against the components of the v_i, they look almost like white noise. Anything that later *looks at* the embedding then cannot generalise from it. So the reduction in (2) is balanced against keeping each output simple. Specifically:

   - Complexity is measured by *degree in the components of the* v_i, not by the size of the expression in d's and e's. Each d has degree 2 and each e has degree k. A finer measure is the multidegree, with one degree for each vector v_i. Every Gram, Plücker and mixed d–e relation is homogeneous, so rewriting through them changes the expression but never the function, and never its degree. The growth in expression size from rewriting $e\,e\,e\,e$ as d's is therefore a problem of how outputs are displayed, not of how smooth they are.
   - Degree does control smoothness. By a Bernstein-type inequality, a polynomial of degree r restricted to the unit sphere has angular gradient at most r times its maximum modulus there.

- Outputs should be *homogeneous* in L, or better still multi-homogeneous, so that each one scales cleanly with the overall size of L. Splitting an output into its homogeneous parts never harms injectivity, because the parts determine the whole, although it does increase the count. The Bursarial $E_{c,j}$ are homogeneous in L only when $k = 2$.
- A homogeneous output f of degree r can optionally be replaced by $\operatorname{sign}(f)\,|f|^{1/r}$. This is a bijection of $\mathbb{R}$, so injectivity is preserved, and every output then scales linearly with L. The cost is that the output is no longer differentiable where $f = 0$.
- Dense combinations that mix many monomials or degrees, such as generic linear projections, are disfavoured.

4. *Evaluation cost* is a secondary concern and is expected to be modest.

4.2 Reductions that are always safe

Removing an output that is given by an *explicit, verifiable formula* (here, a linear one) in terms of the outputs kept can never harm injectivity, and the formula itself is the proof. The following reductions therefore need no judgement:

- if $P \subseteq G$, omit the e's from w from the outset;
- drop coefficients that vanish identically;
- drop exact duplicates and scalar multiples;
- drop any output in the linear span of those kept. This can be detected formally (in the d's and e's) or as functions of L, where the Gram, Plücker and mixed relations hold automatically.

If the reduced set still separates, then every discarded output is still *determined* by the outputs kept. Equal kept outputs imply equal MS, so a discarded output is a well-defined function of the kept ones. What changes beyond the reductions above is that this function is generally not linear, and not even polynomial. For example, x is determined by x^3 on $\mathbb{R}$, but only through a cube root. Discarded invariants typically satisfy polynomial equations over the kept ones, and injectivity is what selects the correct root. The safety of such a reduction therefore cannot be read off from an identity among the outputs. It needs a certificate valid on every input: either an exact algebraic computation (of the *separating variety*), or an explicit decoding algorithm, with case splits for degenerate inputs, that is proved to recover the orbit of L as in target (B). Tests at random points, such as Jacobian rank or counting preimages, can only propose candidates or reject them.

4.3 Levels of certification

"Certified" means that there is evidence that an output set separates orbits on *every* input (target (B)). That evidence comes in levels of decreasing strength.

Level 0: by construction. Injectivity follows from a general theorem, and nothing needs to be checked for the particular (n, k, G). Example: the unpruned Bursarial embedding (by unique factorisation).

Level 1: explicit identity. A Level 0 set from which outputs have been removed, each removed output being an explicit formula in the kept ones ("0", "= output 7", "$= \frac{1}{2}\,\text{output}_3 - \text{output}_9$"). The code verifies each formula exactly, in rational arithmetic, and injectivity is inherited from Level 0. These are the reductions of Section 4.2.

Level 2: proof of separation. The removed outputs are not formulas of the kept ones, but there is a proof that the kept set separates every input. The proof can take one of three forms:

(a) a theorem proved for a whole family of cases and recorded in these notes, so that the code need only check that the requested (n, k, G) satisfies its hypotheses;

(b) an exact computer-algebra computation for this particular (n, k, G), such as showing that the separating variety is exactly the union of the graphs of the group action. The result is stored as an artefact that can be checked again;

(c) a decoder that reconstructs the orbit of L from the outputs, with a written proof, including case splits for degenerate inputs, that it succeeds on every input. The code implements the decoder, and tests exercise it.

Forms (a) and (c) rest on proofs checked by people, and (b) rests on the correctness of the computer-algebra system. None is formally verified. "Certified" therefore means that a proof exists that we stand behind, not that the proof has been checked by machine.

Level 3: evidence only. Tests at random points pass (full Jacobian rank, correct preimage counts, no confusable pairs found by search), but there is no proof. Generic injectivity is then plausible, but nothing reliable is known about degenerate inputs. Level 3 sets are *uncertified*.

4.4 Composing reductions

Certified reductions (Levels 0–2) compose. Each step leaves a set of outputs that is still injective on MS, so whatever a step discards remains determined, in principle, by what it keeps. The order in which reductions are applied therefore cannot affect correctness. It can only affect what later reductions are able to find: an early, suboptimal reduction may make a better one harder to discover, but it never breaks injectivity. Uncertified (Level 3) candidates carry no such guarantee, and do not compose safely.

Because every step is recorded with its certificate, different orders can be tried from scratch and compared, at least for small cases.

Some reductions prune outputs of a fixed embedding (for example, the safe reductions). Others replace the layer 2 embedding itself by another that is injective by construction (for example, involution packing, below). Later reductions then act on the new outputs, and the chain stays certified.

4.5 A pitfall: pairwise zips do not determine a table

A tempting way to exploit structure is to embed each row of the multiset (viewed as a table whose columns are its members) separately, and then "zip" rows together in pairs, for example by embedding the multiset of complex numbers $u + iv$ formed from two rows. *This is not injective.* Two different tables can agree on every pairwise zip. In the smallest example below, the columns are the members:

$$\begin{array}{rcccc} & A & A & B & B \\ T: & X & Y & X & Y \\ & u & v & v & u \end{array} \qquad \begin{array}{rcccc} & A & A & B & B \\ T': & X & Y & X & Y \\ & v & u & u & v \end{array}$$

Every pair of rows has the same multiset of column pairs in T as in T', but (A, X, u) is a column of T and not of T'. A less symmetric example is

$$\begin{array}{rccccccc} & A & A & B & B & B & C & C \\ T: & X & Y & X & Y & Z & Y & Z \\ & v & w & w & v & u & u & w \end{array} \qquad \begin{array}{rccccccc} & A & A & B & B & B & C & C \\ T': & X & Y & X & Y & Z & Y & Z \\ & w & v & v & u & w & w & u \end{array}$$

This counterexample stopped the development of the "lumped multiset zip" idea in the Set Encodings Redux notes.

MS is not a general table: it is a G-orbit. In an orbit, the values of a slot orbit appear equally often, which neither example above respects. But on degenerate inputs, coinciding values merge those counts into unequal ones. So orbit structure cannot be assumed to rescue pairwise (or r-wise) zips. Any scheme built on partial zips needs a proof, valid on every input, that uses the orbit structure.

5 Code interface (draft for discussion)

This section specifies what code in Echinocoder should do, before any code is written. It is intended to be iterated on.

5.1 Two phases: build once, evaluate many times

Build (once, may be slow). The user supplies a specification and receives an embedding function together with its metadata:

```
embed_fn, meta = build_invariant_embedding(k, group_spec, options)
```

- `k`: the dimension of each vector.
- `group_spec`: defines n, the labels, and G. G is a direct product of symmetric groups, one for each *species*, for example $((a, b), (p))$ for $S(2) \times S(1)$, together with a flag for whether P is included. This matches symatom's existing `VectorType/Context`, which already builds w (as `repS`) and the signed-permutation action, and should be reused. Parity is not yet a group element in symatom. This class of groups is sufficient for now. Arbitrary permutation subgroups are out of scope, but the design should not rule them out.
- `options`: control the construction (see below).

Symbolic work happens only at build time: constructing w and G, forming the embedding, applying reductions, and recording provenance.

Evaluate (many times, must be cheap). The target framework is PyTorch, as used by the IWPC package (`https://github.com/jjhwilkinson/iwpc`, which depends on `torch` and `lightning`). `embed_fn` is therefore a `torch.nn.Module`:

```
Y = embed_fn(X)      # X.shape == (..., n, k)  ->  Y.shape == (..., D)
```

- Any leading batch dimensions ("`...`") are allowed, and may differ from call to call. n and k are fixed at build time, and inputs with any other trailing shape are rejected.
- Along the n axis, `X[..., i, :]` is the vector with the i-th label of `group_spec`. That ordering convention is part of the specification.
- Internally the pipeline is: `X` $\rightarrow$ the d's and e's (w, shape $(\ldots, m)$) $\rightarrow$ the polynomials chosen at build time $\rightarrow$ optional root transform $\rightarrow$ `Y`. Only tensor operations are used, so autograd works throughout. No group elements are enumerated at evaluation time. Any constants are registered as buffers, so that `.to(device)` and `state_dict()` behave as usual.
- `Y` must separate orbits for *every* input, and must be differentiable almost everywhere, so that it can be used for backprop. Differentiability may fail at corners: for example, the root transform is not differentiable where $f = 0$. Its gradient there must be defined explicitly (finite, rather than `inf` or `nan`).

- *IWPC compatibility, without depending on IWPC.* Echinocoder must not have a hard dependency on IWPC. It should, however, fit naturally into IWPC, using the same packages (`torch` and `numpy`) and the same conventions as IWPC's `Encoding` base class:
  - an `nn.Module` whose `input_shape` and `output_shape` are registered as integer buffers, and which acts on tensors of shape `(..., *input_shape)`;
  - `input_shape` $= (n, k)$ and `output_shape` $= D$;
  - a flattened variant with `input_shape` $= nk$ (vector-major: component c of v_i at position $ik + c$), because IWPC's `&` concatenation needs vector inputs;
  - NumPy-style docstrings, as in IWPC.

  Wrapping the module as an IWPC `Encoding` should then be a matter of a few lines. Echinocoder may ship such an adaptor, provided IWPC is imported only when the adaptor is used, so that nothing else in Echinocoder needs IWPC installed.
- NumPy inputs may be accepted for convenience, for example in tests and documentation, by conversion to tensors.

5.2 Options (initial list)

- `embedder`: which multiset embedding to use (layer 2), chosen from a registry so that others can be added. Currently `"bursar"` (Section 3.1) and `"bursar_involution"` (involution packing, Section 3.3), both Level 0.
- `reductions`: which reductions to apply, as an ordered list or a level. Early versions may support only `none`. Next would come the safe reductions of Section 4 (zero, duplicate, scalar multiple, linear span, which are Level 1; and dropping the e's when $P \subseteq G$, which is Level 0 for the smaller w, because the d's alone determine L up to $O(k)$). Level 2 reductions are applied only if the corresponding proof, artefact or decoder exists in the code. The main builder never returns a Level 3 set. See Section 4.3.
- `homogenise`: split outputs into homogeneous (or multi-homogeneous) parts in L.
- `root_normalise`: apply $\operatorname{sign}(f)|f|^{1/r}$ to each homogeneous output of degree r.
- `field`: $\mathbb{R}$ initially. Whether to support $\mathbb{C}$ is still open.
- `dtype`: the precision of evaluation, `float32` by default or `float64`. It affects only the evaluator. The build-time symbolic work is exact, with integer or rational coefficients. In keeping with the usual `torch` conventions, the module can also be converted afterwards with `.to(dtype)`.

An embedding that works in `float64` but not in `float32` is already too fragile to be useful. Keeping outputs of low degree and low noise (objective 3 of Section 4) is partly what makes `float32` sufficient, so the designs should aim to work in `float32`. As a diagnostic, the build could compare `float32` and `float64` evaluations on random inputs and record the discrepancy for each output in `meta`.

5.3 Metadata

`meta` is for documentation and auditing only. It is never used in evaluation. It should be serialisable (for example as JSON), and should contain or allow generation of:

- the spec and options it was built from, plus the Echinocoder version or commit, so that the embedding can be rebuilt and checked;

- D, together with the slot order of w;
- for each output: its symbolic formula (see below), its degree and multidegree in L, whether it is homogeneous, any transform applied, and its provenance (for example, "$E_{2,12}$ of the Bursarial embedding");
- a log of the reductions: each discarded output, the reason, and the evidence. Examples are "identically zero", "equal to output i", "$= \sum_i \lambda_i$ (output i)", or a named certificate;
- the *certification level* (Section 4.3) and what it rests on: the theorem, the identities, the stored artefact, or the decoder together with its proof;
- a LaTeX table generator for the outputs.

Symbolic form of the outputs. Evaluation works from the components of L, but the symbolic description never expands below the d's and e's. These are its leaves. Nobody reading `meta` needs to see $EF - p_xq_x - p_yq_y - p_zq_z$ in place of a dot product. Degrees in L are still reported, counting 2 for each d and k for each e.

The description may also use *nested definitions*, naming intermediate quantities once and referring to them by name. For example,

$$\mathrm{enc}[23] = f\big(h(w)\big) + g\big(h(w)\big), \qquad \text{with } h(w) = \dots$$

Natural intermediates are $q_g(x)$, the elementary symmetric polynomials E_c and their coefficients $E_{c,j}$, and orbit sums. This applies to both the machine-readable form (a set of named definitions, such as a DAG of expressions) and the LaTeX output (a table of outputs plus a table of definitions). Its purpose is to keep each displayed formula short. Whether to use nesting, and how deeply, is a presentation choice. It does not affect evaluation.

5.4 Placement and reuse

The code lives in a new Echinocoder subpackage next to symatom and symcoder. It should not inherit from the existing `MultisetEmbedder`, which works on one input at a time and has no group specification.

symatom already builds something close to w (`repS`) and the signed-permutation action of G on it, and it should be reused where it matches. It may do exactly what is needed here, or something subtly different. Before any part of it is relied on, its behaviour must be checked against the definitions of Section 1, in particular:

- which atoms appear: all self-dots, cross-dots and e's, and nothing else;
- the canonical index order, and the sign conventions for e's under relabelling;
- parity. symatom's `Atom.sign` is known to conflate the sign from permuting arguments with the sign from spatial parity. This was harmless while there was no parity operation, but P is now being added.

Tests should compare the w and $g \cdot w$ obtained from symatom with an independent, direct implementation of Section 1 on small cases. symcoder's encoders are numerical, mostly based on sorting, and are not Bursarial. So the main thing to reuse there is perhaps its design experience rather than its code.

5.5 Scale

The initial goal is $k \leq 5$ and $n \leq 10$. Larger would be welcome. The goal has strong consequences. For a single species with $n = 10$, $k = 5$:

$$m = \binom{10}{2} + 10 + \binom{10}{5} = 307, \qquad |G| = 10! \approx 3.6 \times 10^6 \ (\times 2 \text{ with } P), \qquad D_{\text{Bursar}} \approx 2 \times 10^{15},$$

whereas the quotient dimension, which is the lower bound on the number of outputs, is only $nk - \frac{1}{2}k(k-1) = 40$. Hence:

- The unpruned Bursarial embedding is feasible only for small $|G|$. There it serves as a reference baseline, against which reductions can be validated.
- For large G, the build cannot enumerate G, and cannot first form the unpruned output and then prune it. It must construct the reduced outputs directly. One way is to work with orbits of monomials in w (orbit sums), described by the pattern of their labels rather than by listing group elements. symatom's orbit enumerators may help here.
- Evaluation should use tensor contractions, for example $\sum_{i,j} d_{ij}^2 = \operatorname{tr}(\mathbf{d}^2)$, rather than expanded lists of terms, whose length grows with orbit size.

Even $k \leq 5$, $n \leq 10$ may prove too big. It is a goal, not a requirement.

5.6 Uncertified (Level 3) candidates

The main builder returns only sets at Levels 0–2. Level 3 candidates are useful for research, for example to see how small output sets can get and to suggest which sets are worth proving. They are available only through a separate, explicitly named research entry point, *not* through a flag or option on the main builder, so that they cannot be picked up by accident. Their `meta` states that they are uncertified (Level 3), and lists the evidence collected.

6 Exploration log

This section records what has been observed by running the code on very simple cases. It also lists lines of enquiry those observations suggest. The lines of enquiry are *not* a plan. They are unordered, none is singled out as the next thing to investigate, and other ideas are expected to be added and compared with them.

6.1 Tools and outputs

- `symembed/explore/simple_cases.py` writes each case out in full (`docs/explorations/simple_cases.pdf`, with every expansion in `simple_cases_full.txt`). It shows the slots of w, the action of G, every $E_{c,j}$ expanded in the d's and e's, and the effect of the safe reductions.
- The polynomial-closure tool, `symembed/explore/polynomial_closure.py` (report: `docs/explorations/polynomial_closure.txt`), visits outputs in order of c, then j. It keeps an output unless it is an exact polynomial in outputs already kept, and then retries each kept output against the others ("backward pruning"). Membership is tested either formally in w or as functions of L, where the Gram and Plücker relations hold automatically. Identities are found modulo four primes, reconstructed by the Chinese remainder theorem, and verified exactly. Every kept set is therefore a Level 1 certified separating set: each Bursarial output is an explicit polynomial in it. Each case is run for both the plain and the involution-packed embedding.

6.2 Observations

The table gives the number of outputs in each case (all without parity).

case	Bursarial	linear	poly. in w	poly. in L	quot. dim.
$S(2)$, (a,b), $k=1$	14		6	2	2
$S(3)$, (a,b,c), $k=1$	174		21	5	3
$S(2)\times S(1)$, (a,b,p), $k=1$	26		12	3	3
$S(2)$, (a,b), $k=2$	11	8	5	4	3
$S(3)$, (a,b,c), $k=2$	174	155	32	13	5
$S(2)\times S(1)$, (a,b,p), $k=2$	26	22	15	11	5

1. *Polynomial identities do much more than linear ones*, and allowing the Gram and Plücker relations (working in L rather than formally in w) does much more again.

2. *Low c suffices for generation.* For $S(3)$ every kept output has $c \le 4$ ($k=2$) or $c \le 3$ ($k=1$), although the Bursarial embedding goes up to $c = |G| = 6$. All higher-c outputs are polynomials in lower ones.

3. *For $k=1$* (the classical symmetric-polynomial setting), the count reaches the quotient dimension for $S(2)$ and $S(2)\times S(1)$. For $S(3)$ it stops at 5. The search can only keep actual Bursarial outputs, and it is greedy, so this says nothing about how small a set could be.

4. *Polynomial closure gives generating sets, not minimal separating sets.* For $k=2$ the counts stay well above the quotient dimension (11 against 5, and 13 against 5).

5. *$S(2)$, $k=2$, as a zip.* Here

$$\mathrm{MS} = \{(d_{aa}, d_{bb}, d_{ab}, e_{ab}),\ (d_{bb}, d_{aa}, d_{ab}, -e_{ab})\}.$$

   Store the invariant d_{ab} as it is, and zip $u+iv$ with $(u,v) = (d_{aa}, e_{ab})$ in one member and $(d_{bb}, -e_{ab})$ in the other. This gives $d_{aa}+d_{bb}$ (the imaginary part vanishes) and $(d_{aa}d_{bb} + e_{ab}^2) + i\,e_{ab}(d_{bb}-d_{aa})$: 4 reals in all. These coincide with the 4 kept by the polynomial closure, up to a polynomial change of variables. One of the identities needs the Gram relation $e_{ab}^2 = d_{aa}d_{bb} - d_{ab}^2$. The quotient dimension is 3. For fixed $d_{aa}+d_{bb}$ and d_{ab}, the remaining freedom is a closed loop, so any 3-output scheme would have to mix coordinates nonlinearly.

6. *Complex structure for $k=2$.* Identify each v_i with a complex number z_i. Then

$$\bar{z}_i z_j = d_{ij} + i\,e_{ij},$$

   so the d's and e's are the real and imaginary parts of products that are invariant under rotation.

7. *Reference covariants reach the quotient dimension generically.*

   - For $S(2)\times S(1)$, (a,b,p), $k=2$: put $\alpha = \bar{p}a = d_{ap} + ie_{ap}$ and $\beta = \bar{p}b = d_{bp} + ie_{bp}$. When $p \neq 0$ the orbit is determined by $|p|^2 = d_{pp}$ and the unordered pair $\{\alpha, \beta\}$, and so by d_{pp}, $\alpha+\beta$ and $\alpha\beta$. That is 5 reals, equal to the quotient dimension, each of degree 2 in w.
     - When $p=0$, $\alpha=\beta=0$ carries nothing. Adding the 4 outputs of $S(2)$, $k=2$ for (a,b) covers that stratum, and d_{pp} shows which case applies. That gives 9 outputs that separate every input, by a short decoder argument (Level 2c), against 11 from the polynomial closure.

- For $S(3)$, (a, b, c), $k = 2$: the sum $s = \sum_i z_i$ gives $\alpha_j = \bar{s}z_j = \sum_i (d_{ij} + ie_{ij})$. The elementary symmetric polynomials of the α_j give $1 + 2 + 2 = 5$ reals, equal to the quotient dimension, but only where $s \neq 0$. The stratum $s = 0$ needs something else.
- In both examples, each degenerate stratum on which the reference vanishes costs extra outputs. This is consistent with the expectation that sets separating every input are larger than sets separating only generic inputs.

Involution packing under reduction. The counts below run from outputs before reduction, to those kept by the polynomial closure in w, to those kept by the polynomial closure in L.

case	plain	involution-packed
$S(2)$, $k = 2$	$11 \to 5 \to 4$	$5 \to 5 \to 4$
$S(3)$, $k = 2$	$174 \to 32 \to 13$	$90 \to 58 \to 13$
$S(2) \times S(1)$, $k = 2$	$26 \to 15 \to 11$	$14 \to 12 \to 11$
$S(2)$, $k = 1$	$14 \to 6 \to 2$	$8 \to 6 \to 2$
$S(3)$, $k = 1$	$174 \to 21 \to 5$	$111 \to 24 \to 5$
$S(2) \times S(1)$, $k = 1$	$26 \to 12 \to 3$	$17 \to 11 \to 3$

After full closure in L the two agree in every case. Formally in w the packed embedding is sometimes worse (58 against 32 for $S(3)$, $k = 2$). Packing halves the raw count at no cost, but leaves unchanged what survives the polynomial closure, which is itself only feasible for small cases.

6.3 Possible lines of enquiry (unordered, not exhaustive)

- Extending the safe reductions from linear to polynomial identities (Level 1). On the cases above this alone takes, for example, 26 down to 11.
- Reference covariants, for $k = 2$ via the complex structure, with the degenerate strata handled by further covariants or by recursion to smaller cases. Open questions: whether there is an analogue for $k \geq 3$, which has no complex structure, and how the number of strata grows.
- Understanding why low c suffices for generation, and whether there is a useful bound on the c needed in terms of the structure of G.
- Further simple cases for comparison: parity, $k = 3$, and two non-trivial species.
- Complex packing of w: pair the real slots of w into $\lceil m/2 \rceil$ complex numbers, embed the multiset of packed vectors with the complex Bursarial embedding, and write out real and imaginary parts. The raw count changes very little, because the Bursarial size is linear in m. For even m it saves $|G|(|G| - 1)/2$ reals (as already noted in the legacy embedder's docstring), which is a fraction of about $1/m$. For odd m, padding with zero costs $|G|$ reals. Any real benefit would have to show up in what survives reduction, and that may depend on the pairing (the order of the slots). A factor of 2 would need an element of G acting on the packed vector as complex conjugation, and without parity no permutation other than the identity does so.
- Involution packing, which generalises the $z \mapsto i\bar{z}$ trick and does not need parity. Choose any involution $g_0 \in G$. It splits w into a $+1$ part and a -1 part. A slot pair (s, t) swapped by g_0 contributes $w_s + w_t$ and $w_s - w_t$, with the signs adjusted as required, and a fixed slot contributes to the part given by its sign. Pack the $+1$ coordinates as real parts and the -1 coordinates as imaginary parts, padding with zeros if the counts differ. Then g_0 acts on every packed vector as complex conjugation. The multiset of packed vectors is

therefore closed under conjugation, so $\prod_g (y + q_g(x))$ has real coefficients, and only those need storing.

- This is Level 0: the complex Bursarial embedding is injective by unique factorisation, and the packing is a bijection.
- The number of packed slots is $m'' = (m + |\operatorname{tr} g_0|)/2$, so the involution with smallest $|\operatorname{tr}|$ is best.
- Raw counts fall to roughly half: $11 \to 5$ for $S(2)$, $k = 2$; $174 \to 90$ for $S(3)$, $k = 2$; $26 \to 14$ for $S(2) \times S(1)$, $k = 2$; $138060 \to 79980$ for $S(5)$, $k = 4$; $1884 \to 948$ for $S(3) \times S(2)$, $k = 3$.
- Now implemented (Section 3.3, embedder `bursar_involution`). What survives full polynomial closure is unchanged on the simple cases (see above). Open questions: whether packing changes what survives other reductions, and whether the factor of 2 can be exceeded.
- Going beyond a factor of 2 with two commuting involutions would need an algebra with two independent conjugations. Bicomplex numbers have zero divisors, which breaks unique factorisation, and quaternions do not commute, so the product depends on the order of its factors.